

Lab 01 - The Price of One Request

Google Gemini, gemini-3.5-flash

Pricing: standard paid tier, \$1.50 per 1M input tokens and \$9.00 per 1M output tokens (including thinking tokens). Workload: 2,000 support requests/day = 730,000 requests/year; this is a small realistic queue volume that makes the language difference visible.

1. Prediction versus measurement

Part 1 predicted the multiplier from UTF-8 byte length: Russian 1.92x and Kazakh 2.13x versus English.

Language	Prediction vs EN	Measured request	Measured vs EN	Billed input / output
English	1.00x	100	1.00x	101 / 535
Russian	1.92x	130	1.30x	131 / 698
Kazakh	2.13x	236	2.36x	237 / 1,160

The prediction was directionally correct for Kazakh but overestimated Russian: the measured Kazakh request was 2.36x the English request, while Russian was 1.30x.

2. Annual cost at 2,000 requests/day

Cost/request = (input tokens x \$1.50 + output tokens x \$9.00) / 1,000,000.

Language	Cost/request	Annual cost
English	\$0.00497	\$3,625.55
Russian	\$0.00648	\$4,729.31
Kazakh	\$0.01080	\$7,880.72

3. Production choice for a Kazakh support queue

I would start production with Gemini 3.5 Flash: the measured Kazakh response was fluent, in the requested language, and correctly refused to invent missing contract data. Its measured cost is about \$7,881/year at the chosen volume, or 2.17x English, so I would add human escalation/quality checks for answers that go beyond the supplied documents. Before a final commitment, I would A/B-test a cheaper Flash-Lite model on a Kazakh evaluation set; it was not measured in this lab.

4. Cost-saving lever not used

I would enable context caching for the repeated system prompt, which this lab did not use, so the same instructions are not charged as fresh input on every request.

Pricing source: Google Gemini Developer API pricing, Gemini 3.5 Flash Standard tier, accessed 2026-09-19: ai.google.dev/gemini-api/docs/pricing.

AI-use declaration

I used Codex to inspect the laboratory repository and instructions, adapt the measurement script from the original provider to the Google Gemini API, troubleshoot SDK/API errors, calculate the annual-cost table, and format this submission. I used the Gemini API with gemini-3.5-flash to measure token counts and generate the English, Russian, and Kazakh test responses.

I ran the scripts myself, checked the dry run and the successful measurements, and used the returned token-usage metadata for the numbers above; the measurements were not taken from an internet estimate. I did not include or disclose my API key. The final workload assumption, model recommendation, interpretation of quality, and wording of this declaration are my responsibility.